

Advanced Data Management

Course Presentation

*Corso di laurea magistrale
INGEGNERIA INFORMATICA
Dipartimento di Ingegneria Gestionale, dell'Informazione e della Produzione
University of Bergamo*

Main Aspects

- Course Title: **Advanced Data Management & Laboratory**
- 9\6 Credit Course
- **Lecturers:**
 - Prof. Stefano Paraboschi (parabosc@unibg.it)
 - Prof. Domenico Fabio Savo (domenicofabio.savo@unibg.it)
- **Website:** <https://cs.unibg.it/savo/didattica/adm/adm2425/index.html>
- **MS Teams:** The link to join the group is available on the course website

Course description

The course is structured in two modules:

- **Module 1 (6 CFU)**

- **Lecturers:** [Prof. Domenico Fabio Savo](#)
- **Description:** This module presents an overview of the research work carried out in the area of database theory and data integration, with emphasis on the theoretical aspects relevant to the development of information integration solutions. Special attention will be devoted to the architectures for data integration systems, modeling an information integration application, data exchange, query answering in data integration, and semantic technologies for data management.

- **Module 2 (3 CFU)**

- **Lecturers:** [Prof. Stefano Paraboschi](#)
- **Description:** The second module (of 3 CFU) focuses on technologies for managing data sets whose size exceeds the processing capacity of conventional database systems.

Timetable

- **Tuesday**, 8:30 – 10:30, Room B004 (**Prof. Savo**)
- **Wednesday**, 16:00 – 19:00, Room B102 (**Prof. Savo**)
- **Friday**, 10:30 – 13:30, Room B-106 (**Prof. Paraboschi**)

(Sometimes we will switch time slots)

It is recommended to consult the online timetable to check for any changes made to the standard schedule and the course's **MS Teams Group**

Office Hours

- Prof. Stefano Paraboschi:
 - **When:** Wednesday, 17:30 – 19:30
 - **Where:** B building, 2nd floor, office #226
- Prof. Domenico Fabio Savo:
 - **When:** Tuesday, 10:30 – 12:30
 - **Where:** B building, 2nd floor, office #225

Please, contact the teachers by email to arrange a meeting.

Exam Structure

The exam is structured in two parts:

1. Study and presentation of a research paper chosen by the student and approved by the teacher.
 2. A written test with questions and exercises about the topics covered in the course.
-

Course Material

Textbook:

- Textbook in Italian: Paolo Atzeni, Stefano Ceri, Piero Fraternali, Stefano Paraboschi, Riccardo Torlone "Basi di dati", Quinta edizione, McGraw-Hill
- Textbook in English: "Database systems", McGraw-Hill. The textbook is freely available in PDF from the Website.

Slides:

Slides and other additional teaching material will be uploaded on MS Teams as the course progresses.